

Lesson 08 · Derivatives of exponentials and logarithms

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q1 · Rank 1 / Foundation

Write down the derivative of $y = e^{6x}$.

☐ A e^{6x}

☐ B $6e^x$

☐ C $6e^{6x}$

☐ D $e^{6x}/6$

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Q2 · Rank 1 / Foundation

Write down the derivative of $y = e^{-342x}$.

☐ A $342e^{-342x}$

☐ B e^{-342x}

☐ C $-342e^x$

☐ D $-342e^{-342x}$

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Q3 · Rank 1 / Foundation

Write down the derivative of $y = \ln(3x)$, where $x > 0$.

☐ A $1/x$

☐ B $3/x$

☐ C $1/(3x)$

☐ D $\ln 3$

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Q4 · Rank 2 / Core

Write down the derivative of $y = 2e^{-x} + 4e^x$.

☐ A $2e^{-x} + 4e^x$

☐ B $-2e^{-x} + 4e^x$

☐ C $-2e^{-x} - 4e^x$

☐ D $-e^{-x} + e^x$

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Q5 · Rank 2 / Core

Write down the derivative of $y = 10e^{4x} - 2x^2 + 7$.

☐ A $10e^{4x} - 4x$

☐ B $40e^{4x} - 4x + 7$

☐ C $40e^{4x} - 4x$

☐ D $40e^x - 4x$

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Q6 · Rank 2 / Core

Write down the derivative of $y = \ln(-13x)$, where $x < 0$.

☐ A $-13/x$

☐ B $-1/x$

☐ C $1/(-13x)$

☐ D $1/x$

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Q7 · Rank 3 / Application

If USD 4000 is saved in an account offering a return of 4% compounded continuously, the future value, S , after t years is given by $S = 4000e^{0.04t}$. Write down an expression for dS/dt and hence find the exact value of the rate of growth after five years.

☐ A $160e^{0.2}$

☐ B $4000e^{0.2}$

☐ C $0.04e^{0.2}$

☐ D $160e^{0.04}$

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Q8 · Rank 3 / Application

If USD 4000 is saved in an account offering a return of 4% compounded continuously, the future value, S , after t years is given by $S = 4000e^{0.04t}$. Calculate S at $t = 5$ and $t = 5.01$ and hence estimate the rate of growth at $t = 5$. Use unrounded intermediate balances; round the rate to two decimal places.

☐ A 1.95

☐ B 195.46

☐ C 195.42

☐ D 4885.61

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Q9 · Rank 3 / Application

Use the product rule to differentiate $y = x \ln x$. Assume $x > 0$.

☐ A 1

☐ B $x + \ln x$

☐ C $\ln x + 1$

☐ D $\ln x + 1/x$

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Q10 · Rank 4 / Challenge · OPTIONAL

Use the chain rule to differentiate $y = \ln(x^4 + 3x^2)$. Assume $x \neq 0$.

☐ A $1/(x^4 + 3x^2)$

☐ B $4x^3 + 6x$

☐ C $1/x$

☐ D $\frac{4x^3+6x}{x^4+3x^2}$

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